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IT4010 Research Project – 2026 Jul

Topic Assessment Form

V2.2

Project ID: J26-SE-325

1. Topic (12 words max)

An Integrated AI-Driven Smart Finance Platform for Portfolio Optimization, Fraud Detection, Blockchain Aditability & Explainable Agentic Assistance

2. Research group the project belongs to

Centre of Excellence for AI (CEAI)

3. Specialization of the project belongs to

Software Engineering

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The problem affects a broad range of stakeholders across the digital finance ecosystem. The primary affected group is the retail investor/non-expert user, individuals who hold savings or investments on digital platforms but lack the financial literacy, time, or expertise to manage portfolio risk, respond to market volatility, or interpret algorithmic decisions made on their behalf. A secondary group includes fraud and compliance analysts within FinTech institutions, who currently rely on fragmented, offline-validated fraud detection tools that are not stress-tested against live, adversarial conditions. Regulators and auditors form a third group, requiring tamper-evident, privacy-preserving transaction trails that current loosely-coupled blockchain-AI systems cannot reliably provide. Representative user personas include: (1) the "Time-Poor Retail Investor" who needs instant, loss-minimized liquidity without manually rebalancing a portfolio; (2) the "Fraud Analyst" who needs real-time, explainable anomaly signals coupled to enforcement actions rather than static post-hoc reports; (3) the "Compliance Officer/Regulator" who needs an auditable, privacy-preserving record of AI-driven financial decisions; and (4) the "Non-Expert End User" who needs natural-language, trustworthy explanations of automated decisions rather than expert-oriented model outputs.

Impact of the Problem

1. Social Impact: Lack of accessible, explainable financial automation

disproportionately disadvantages non-expert users, widening the gap between financially literate and illiterate populations in capital markets participation.

2. Technical Impact: Fragmented systems (siloed forecasting, fraud detection, and audit layers) create integration debt, inconsistent latency guarantees, and an inability to evaluate AI components under real operational stress, including adversarial conditions.
3. Economic Impact: Unmitigated fraud and poorly timed forced liquidations cause direct financial loss to users and reputational/regulatory cost to platforms; an integrated liquidity-aware optimizer and real-time fraud gateway directly reduce realized slippage and fraud losses.
4. Ethical Impact: Opaque algorithmic decision-making in finance raises fairness and accountability concerns, particularly where automated decisions affect users' access to their own funds or flag them as suspicious without recourse or explanation.
5. Legal Impact: Financial regulators increasingly mandate explainability, auditability, and bias mitigation (e.g., under emerging AI and data-protection regulations); platforms lacking tamper-evident audit trails and human-override governance face compliance exposure.
6. Environmental Impact: Blockchain-based auditability introduces a computational/energy cost consideration; the project mitigates this by favoring permissioned/testnet chains and lightweight commitment schemes (e.g., Merkle anchoring) over energy-intensive public consensus mechanisms.

FoC Research Cluster and Industry Vertical/s

1. Research Cluster: Centre of Excellence for AI (CEAI) – Artificial Intelligence & Data Science (Machine Learning, Agentic AI/LLMs, Explainable AI, and Applied Cybersecurity), with secondary alignment to Distributed Systems and Blockchain Technologies
2. Industry Vertical(s): Financial Technology (FinTech); Banking, Financial Services & Insurance (BFSI); RegTech (Regulatory Technology).

Alignment with Sustainable Development Goals (SDGs)

1. SDG 8 - Decent Work and Economic Growth: Promotes inclusive, secure access to financial services and protects user assets through real-time fraud prevention.
2. SDG 9 - Industry, Innovation and Infrastructure: Advances resilient, transparent digital financial infrastructure through AI–blockchain integration and responsible-AI engineering.
3. SDG 10 - Reduced Inequalities: Improves financial inclusion by giving non-expert retail users access to explainable, automated portfolio and

liquidity management previously available only to institutional investors.

4. SDG 16 - Peace, Justice and Strong Institutions: Strengthens transparency and accountability in financial decision-making through tamper-evident blockchain auditability and governance mechanisms for human oversight

6. **Brief description of Existing Research and Systems (300 words max)**

Commercial robo-advisors (e.g., Betterment, Wealthfront-style platforms) automate long-term portfolio allocation using mean-variance optimization but offer no instant, loss-minimized withdrawal mechanism. Production fraud systems in FinTech rely heavily on rule-based engines and supervised classifiers (e.g., XGBoost, isolation forests) evaluated on benchmark datasets such as PaySim, IEEE-CIS, and Kaggle Credit Card Fraud. Blockchain-based audit platforms (built on Hyperledger Fabric and Ethereum) exist for trade settlement and KYC record-keeping, while early-stage agentic AI assistants are emerging in customer support and compliance contexts, though without empirical trust evaluation.

Academic research has advanced ML-based forecasting and reinforcement-learning-driven portfolio optimization, hybrid LSTM+GNN and RGCN-based fraud detection (reporting near-perfect AUROC on relational benchmarks), behavioral biometrics for anomaly signals, and one-class methods (OCSVM, autoencoders) for zero-day fraud. On the blockchain side, research has explored Merkle-tree commitments, Zero-Knowledge Proofs, homomorphic encryption, and federated learning for privacy-preserving auditability. Explainable AI research (SHAP, LIME) has improved feature-level model transparency for financial decisions.

Despite these advances, no platform unifies them. Liquidity-aware withdrawal planning has not been operationalized as a real-time, user-facing service. Fraud models are validated offline, with no integration into a live security gateway and no systematic testing against adversarial evasion (camouflage, slow-drift, structuring attacks). Blockchain-AI integrations remain loosely coupled, lacking a formal bridge translating probabilistic AI outputs into deterministic smart-contract logic, comparative privacy-mechanism benchmarking, model-versioning provenance, and on-chain human-override governance. Finally, XAI methods remain expert-oriented, and localized agentic assistants have not been empirically evaluated for retail user trust and perceived control.

7. **Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).**

The proposed platform directly closes the four gap clusters identified in Section 6. The liquidity-aware forecasting and optimization engine (Component 1) operationalizes instant, loss-minimized withdrawal planning as a live microservice, something existing robo-advisors do not support. The hybrid LSTM+GNN fraud engine (Component 2) is embedded directly into a deterministic security gateway, so anomaly scores actively modulate rate-limiting and authentication in real time, and is continuously tested against adversarial evasion rather than static benchmarks alone. The AI-to-smart-contract bridge (Component 3) tightly couples off-chain AI outputs with on-chain enforcement within a single transaction lifecycle, using Merkle/ZKP-based privacy-preserving anchoring and model-versioning provenance, resolving the loose-coupling and governance gaps in current blockchain-AI research. The localized agentic LLM assistant (Component 4) translates SHAP/LIME outputs into natural-language, retail-friendly explanations within a Trust Panel, addressing the expert-versus-retail-user explainability mismatch and enabling empirical measurement of user trust and perceived control.

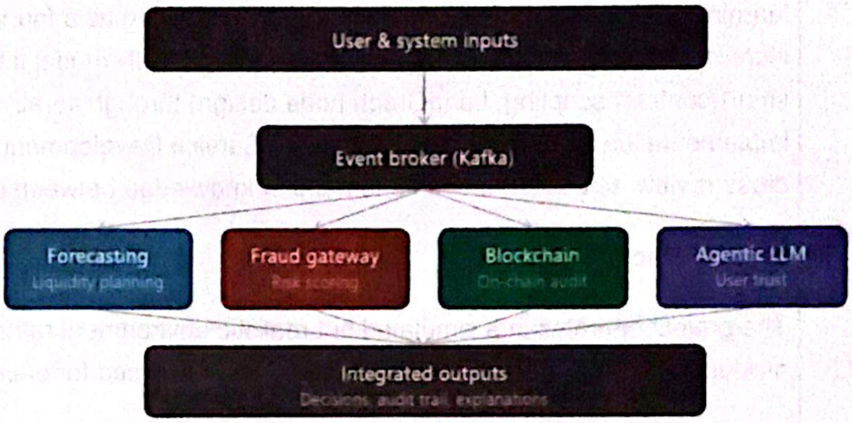
System Perspective: Input – Process – Output

Input: Market/OHLCV data, news and sentiment feeds, simulated user transactions and account balances, behavioral biometrics and device fingerprints, user requests (withdrawals, portfolio queries, natural-language questions), and identity/KYC data.

Process: (1) Multivariate forecasting (TimesFM + LSTM/MLP) and multi-objective, liquidity-aware portfolio optimization (MOEA/D); (2) Real-time transaction graph construction and dual-stream LSTM+GNN fraud scoring, coupled with deterministic gateway enforcement and online learning from analyst feedback; (3) Hashing and privacy-preserving anchoring (ZKP/Merkle) of AI decisions, smart-contract execution, and model-versioned provenance logging; (4) LangGraph-orchestrated agentic reasoning, SHAP/LIME-based explanation generation, and responsible-AI guardrail enforcement (fairness checks, consent flows).

Output: Optimized portfolio allocations and instant-withdrawal plans with minimized slippage; real-time fraud/risk scores with automated enforcement actions (step-up authentication, rate limiting); a tamper-evident, queryable on-chain audit trail and compliance dashboard; and natural-language explanations delivered through a Trust Panel that supports user confirmation, rollback, and oversight of automated actions.

All four subsystems communicate asynchronously through a centralized event broker (e.g., Kafka/EventBridge), ensuring each component can be developed, evaluated, and scaled independently while remaining part of one coherent, auditable decision pipeline.



8. **Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)**

Gaining Specialized Domain Expertise

Each member builds the domain expertise required for their component primarily through structured literature review of the peer-reviewed and industry sources already surveyed in this proposal, covering robo-advisor and liquidity-constrained portfolio theory (Member 1), GNN/RGCN-based fraud detection and adversarial robustness research (Member 2), blockchain-AI integration, ZKP and Merkle-based auditability literature (Member 3), and XAI (SHAP/LIME) and responsible-AI/TRiSM frameworks (Member 4). This is supplemented by academic supervisor guidance, consultation of regulatory and FinTech industry publications, and engagement with open-source community documentation and reference implementations in each domain.

Acquiring Technical Knowledge and Skills

The technical stack required (TimesFM, LSTM/GNN architectures, MOEA/D optimization, LangGraph orchestration, Hyperledger/Ethereum smart contract development, ZKP libraries) is acquired through a combination of official framework documentation, open-source tutorials, online courses, and hands-on prototyping. Members leverage existing coursework in machine learning, distributed systems, and software engineering as a foundation, and incrementally build component-specific skills (e.g., GNN model training, smart-contract scripting, LangGraph node design) through iterative implementation during Phase 3 (Model and Service Development), with peer cross-review across components to transfer knowledge between members.

Data Collection

The project operates in a simulated but realistic environment rather than a live production setting, which determines how data is sourced for each component:

- *Forecasting/Portfolio (Member 1):* Historical OHLCV price data and news/sentiment data are obtained from public market-data and news APIs, restricted to a limited universe of liquid equities and/or ETFs to keep computation tractable.
- *Fraud Detection (Member 2):* Established public benchmark datasets (e.g., PaySim, IEEE-CIS, Kaggle Credit Card Fraud) are used for training and evaluation, supplemented by simulated transaction streams and synthetic behavioral/biometric signals where real data is unavailable or sensitive.
- *Blockchain (Member 3):* Audit and transaction events are generated synthetically within the platform itself and recorded on a permissioned testnet or local development chain rather than a live mainnet.
- *Agentic LLM (Member 4):* Evaluation data primarily comes from a user study, supplemented by synthetic financial scenarios for explanation generation.

User funds throughout the platform are modeled as simulated platform balances rather than real brokerage holdings, which avoids the need to access live, sensitive financial accounts.

Ethical Clearance

Most components rely on public benchmark datasets, synthetic/simulated data, and testnet blockchain infrastructure, which minimizes direct ethical risk. However, ethical clearance is required for Component 4's empirical user study (RQ4), which involves human participants evaluating the agentic LLM assistant for trust, understanding, and perceived control. This requires standard institutional ethical review covering informed consent, voluntary participation, and data anonymization for any participant feedback or survey data collected. Additional care is taken around fairness and bias evaluation in the fraud detection and credit-relevant models, per the responsible-AI mitigation strategies outlined in Section 13 (Risks, Ethics, and Mitigation).

9. Objectives and Novelty

Main Objective
Design, implement, and evaluate a prototype AI-Driven Smart Finance Platform that unifies portfolio management, liquidity optimization, fraud detection, blockchain-based auditability, and agentic AI assistance within a responsible-AI framework, demonstrating that these capabilities can operate as one coherent, empirically evaluated system rather than as isolated subsystems.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Nivakaran S. IT23259416	Design a liquidity-aware forecasting and portfolio optimization engine that supports instant, loss-minimized user withdrawals alongside long-term return optimization.	Build data ingestion and feature-engineering pipelines (OHLCV, news/sentiment); compute technical indicators (MACD, RSI, MFI, ATR); develop the hybrid TimesFM + LSTM/MLP forecasting engine; implement the multi-objective, liquidity-aware optimizer (MOEA/D, fuzzy genetic algorithm); build the instant-withdrawal/liquidation planning module; expose all services via APIs; containerize and monitor the services.	Operationalizes liquidity-aware withdrawal planning as a real-time, user-facing microservice, jointly handling forecasting, rebalancing, and liquidation planning to minimize slippage and realized loss at the moment of cash-out. Existing robo-advisors optimize for long-term risk-return only and do not address this instant-liquidity gap.

Dushanthini R. IT23150348	Develop a real-time fraud and behavioral anomaly detection engine tightly coupled with a deterministic security enforcement gateway, robust to adversarial evasion	Build streaming feature ingestion and dynamic transaction graph construction; implement the hybrid LSTM autoencoder + GNN dual-stream anomaly classifier; incorporate behavioral biometrics and device fingerprinting; build the security gateway (rate limiting, IP reputation scoring, step-up authentication); implement concept-drift monitoring and adversarial robustness testing (camouflage, slow-drift, structuring attacks); build the analyst feedback dashboard and online learning pipeline.	Couples GNN-based anomaly scores directly to live, deterministic gateway enforcement in real time, and validates the model under active adversarial evasion strategies in a live environment, rather than evaluating fraud detection only as an offline classifier on static benchmark datasets, as nearly all published research currently does.
Abeysekara W.C.S.M IT23225442	Design a privacy-preserving, tamper-evident blockchain audit layer that bridges continuous AI decisions to deterministic on-chain smart contract enforcement.	Develop smart contracts for vault token management and event logging; implement the AI-to-smart-contract bridge translating AI risk/liquidity outputs into on-chain logic; implement privacy-preserving anchoring using ZKPs and/or Merkle tree commitments; build model-versioning and provenance tracking for logged AI explanations; design the	Provides an end-to-end design coupling real-time AI decision commitment with smart contract enforcement within a single transaction lifecycle, including a formal bridge pattern for translating probabilistic AI outputs into deterministic logic and an on-chain governance mechanism for human-override accountability, gaps unaddressed by existing

			loosely-coupled blockchain-AI implementations.
W.V.A.D.K. Chamara IT23163904	Build a localized, explainable agentic LLM assistant, empirically evaluated for its effect on user trust and perceived control, within a responsible-AI governance framework.	Implement the LangGraph-orchestrated agentic assistant backend; build the natural-language explainability layer using SHAP/LIME; implement Responsible-AI guardrails (TRiSM, fairness metrics, bias mitigation); build the Trust Panel with confirmation/rollback workflows; implement consent flows and privacy settings; design and run the user trust evaluation study.	Translates expert-focused explainability outputs (SHAP/LIME) into natural-language, retail-friendly explanations and empirically measures their effect on user trust and perceived control, an evaluation that does not currently exist for localized, responsible-AI-compliant agentic assistants in personal finance.

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10. This section is to be completed by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Data required for the project is available.

Yes	✓	No	

f) Any other comments:

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	Title	First Name	Last Name	Signature
Supervisor*	Mr.	Eishan	Weerasinghe	
Co-Supervisor*	Ms.	Poojani	Gunathilake	
External Supervisor				

Summary of external supervisor's (if any) experience and expertise	

*- At this stage, identifying the Co-Supervisor is compulsory. Students should reach out to a staff member, with guidance from their supervisor, to serve as the co-supervisor for this project.